Dhrumil Bhut

Software Engineer | Backend Systems & AI Infrastructure
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PROFESSIONAL SUMMARY

Backend and AI Engineer with 3+ years building distributed async systems and LLM infrastructure in production. Node.js and TypeScript on the backend, Python for AI infra: RAG pipelines, evaluation tooling and observability systems shipped end to end.

WORK EXPERIENCE

Software Engineer 08/2023 to Present
Zuru Tech India Pvt. Ltd., Ahmedabad, India

- Built the brand asset library end to end from scratch across 20+ brand accounts and 20K+ media assets: nested folders with move, copy and cascading delete, multi-filter retrieval via CTEs that cut query time by 40% and improved API response time by 20%, watermarking with 6 placement options, bulk upload, subtitle support, SKU filtering and role-based access.
- Designed the schema and dual state machine for a multi-party licensor approval workflow from scratch: separate status lifecycles for projects and individual assets, hybrid core columns and JSONB for dynamic per-category fields, bulk asset review actions and role-based write access enforced at the brand level.
- Built the async media publishing pipeline using BullMQ and Redis for distributed job scheduling with deduplication, exponential backoff retries and concurrency limits, and RabbitMQ consumer pipelines with cron schedulers to handle media publishing across Instagram, TikTok and YouTube entirely outside the request lifecycle.
- Built a dedicated webhook ingestion service that receives real-time events from Instagram and TikTok, publishes them into RabbitMQ and fans out to multiple internal consumers independently without coupling to the external API contract.
- Architected serverless media processing on AWS: preview generation via Lambda cut compute costs by ~20% and load times by ~30%. Built an S3 photo caching layer for creator profiles that copies on save and deletes on remove, solving CDN expiry for third-party marketplace photos.
- Designed tiered rate limiting: 200 req/10min general, 200 req/hr per user against Meta's 1,000/hr cap and 10K quota/day for YouTube, using two independent middleware chains with no cross-interference. Implemented Azure AD SSO end to end: access-token validation, email-domain enforcement and handling for disabled and deleted users.

PROJECTS

BehavTest *(Behavioral Regression Testing for AI Applications)*
github.com/dhrumilbhut/behavtest | dhrumilbhut.github.io/behavtest | npmjs.com/package/behavtest

- Open source behavioral regression testing tool for LLM apps, agents and RAG pipelines. Shipped as a product: published on npm, listed on the GitHub Marketplace, CI green on Ubuntu, macOS and Windows, versioned releases and a full documentation site.
- Core engine runs each test case through the real pipeline multiple times and compares against a baseline using statistically honest methods: Wilson confidence intervals, Fisher exact test and case-stratified permutation tests. Reports not significant when the data does not support a conclusion rather than forcing a verdict.
- Ships with RAG scorers (faithfulness at claim and answer level, retrieval with hit/recall/precision/MRR, contextRelevance per chunk), a local dashboard with run comparison and calibration views, a GitHub Action for CI and a fire-and-forget SDK that adds zero latency to the host pipeline.

ContextLens *(Self-Hosted RAG Observability Tool)* github.com/dhrumilbhut/ContextLens | contextlens.dhrumilbhut.com

- Built end to end: Python SDK, FastAPI ingest API, Celery worker for async claim processing, PostgreSQL with pgvector, Redis and a Next.js dashboard. Full Docker Compose deployment with a landing page and documentation site.
- Pipeline decomposes every LLM response into atomic claims, embeds claims and source chunks in a single batched API call, attributes each via cosine similarity and classifies failures as retrieval failures (chunk never fetched) or generation failures (LLM misrepresented a chunk it had). Validated on a real 14-query RAG run: initial 0.75 threshold misclassified paraphrased claims; fixed with a three-band confidence model and separate refusal detection.

PayFlow *(Payment Processing System)* github.com/dhrumilbhut/PayFlow

- Async payment processing system with Node.js, RabbitMQ, PostgreSQL and Redis. Two-layer locking via Redis SET NX and PostgreSQL FOR UPDATE ensures no payment is double-processed under concurrent worker load.
- Circuit breaker for fast-fail on gateway outages, exponential backoff retries via Dead Letter Exchange and a crash-recovery watchdog for zero stuck payments under failure. Deployed with 47 unit and integration tests covering state transitions, race conditions and webhook conflict resolution.

SKILLS

Languages	Python | JavaScript | TypeScript | SQL
Backend & API	Node.js | Express.js | FastAPI | Next.js | AWS Lambda | Azure AD SSO | S3
Queues & Jobs	RabbitMQ | BullMQ | Redis | Dead Letter Exchange | Celery
AI & ML Infrastructure	RAG Pipelines | Embeddings | Vector Search | pgvector | LLM Evaluation | OpenAI API
Databases	PostgreSQL | SQLite | Redis | Query Optimization | Schema Design
Concepts	Distributed Systems | Event-Driven Architecture | Idempotency | Microservices | System Design | Webhooks
Infrastructure & DevOps	Docker | Docker Compose | GitHub Actions | AWS | CI/CD
Testing	Jest | Unit Testing | Integration Testing | Regression Testing | Statistical Testing

EDUCATION

B.E. Computer Science & Engineering 2020 to 2024
Darshan Institute of Engineering and Technology, Rajkot, India CGPA: 8.74